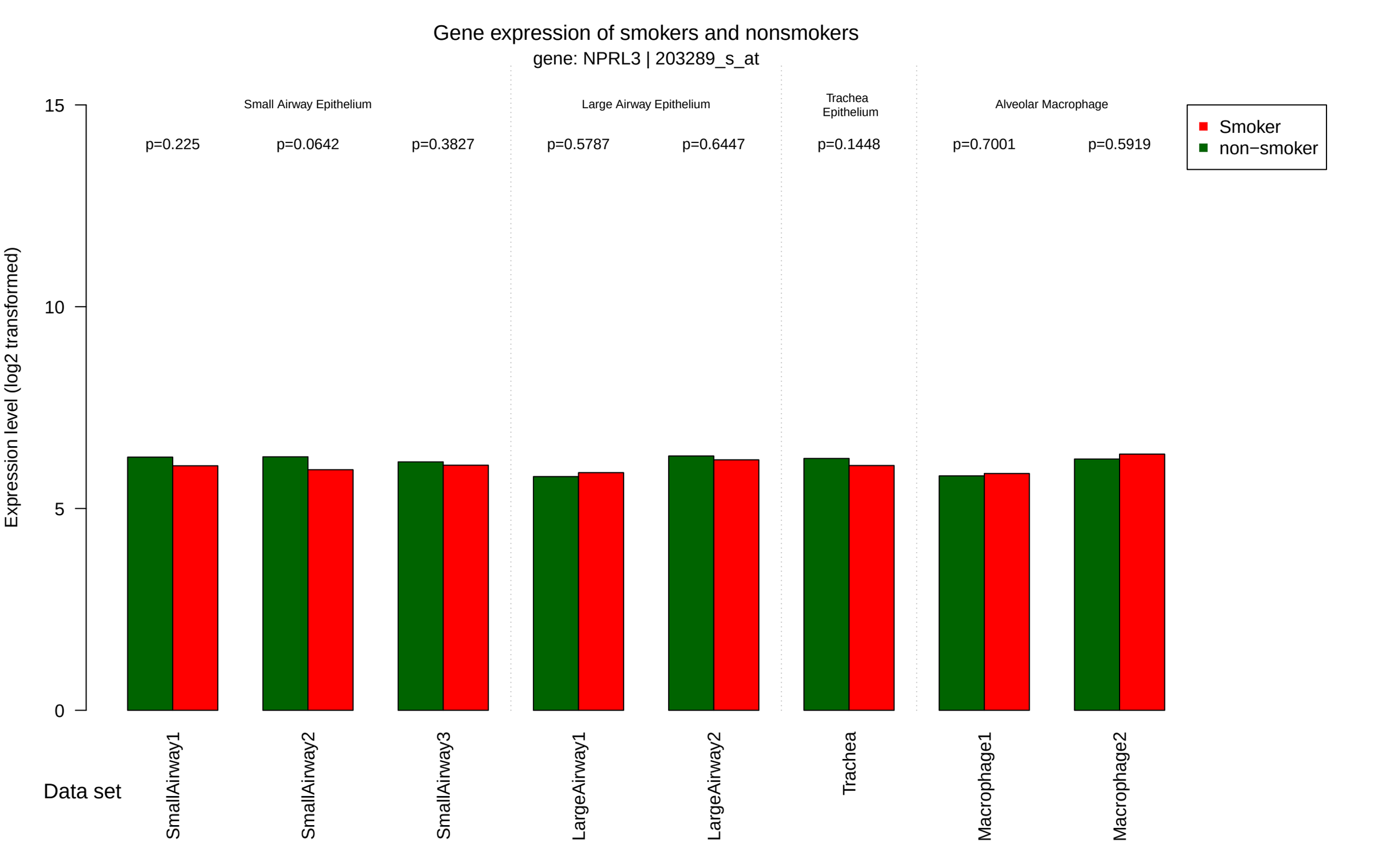
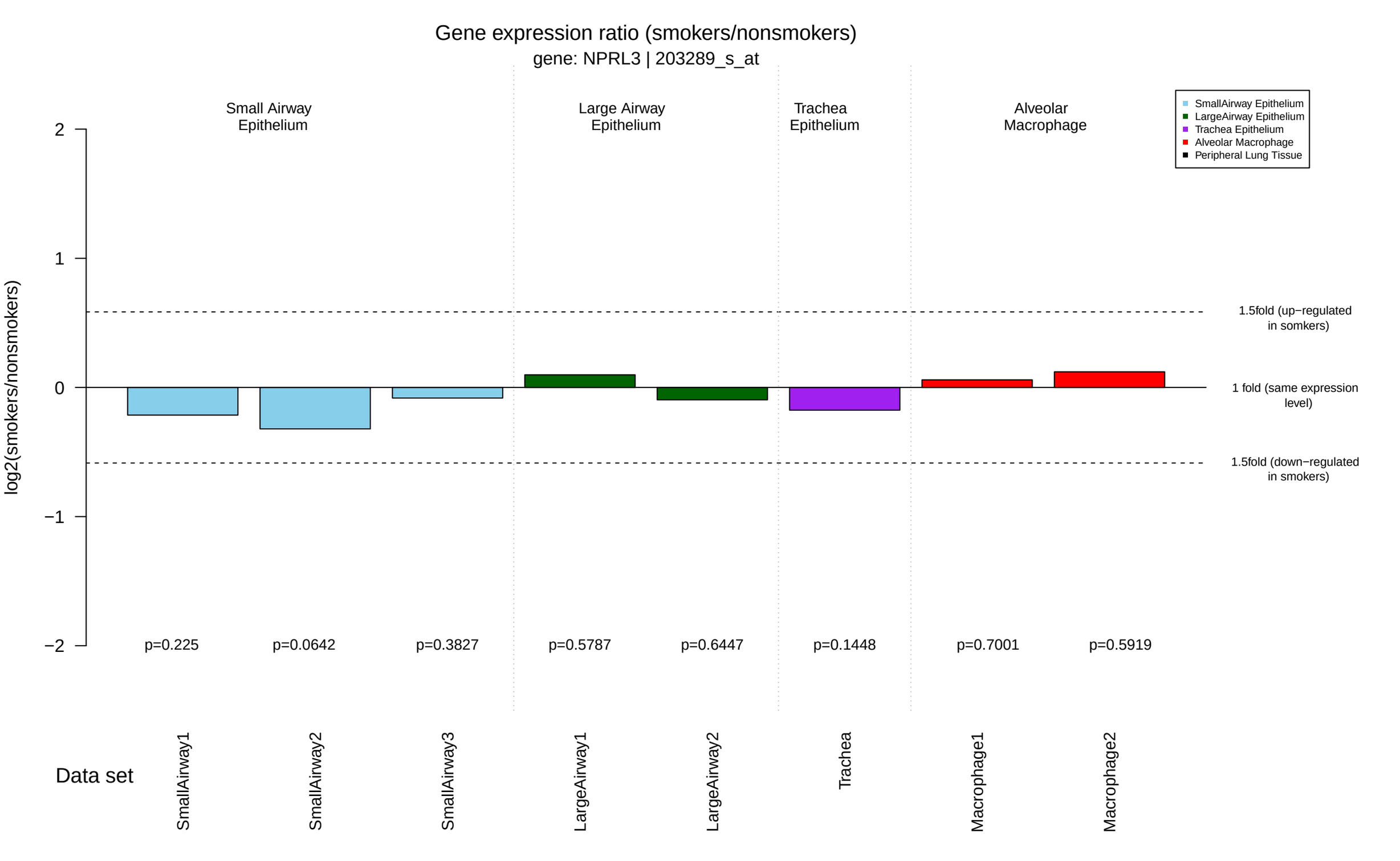
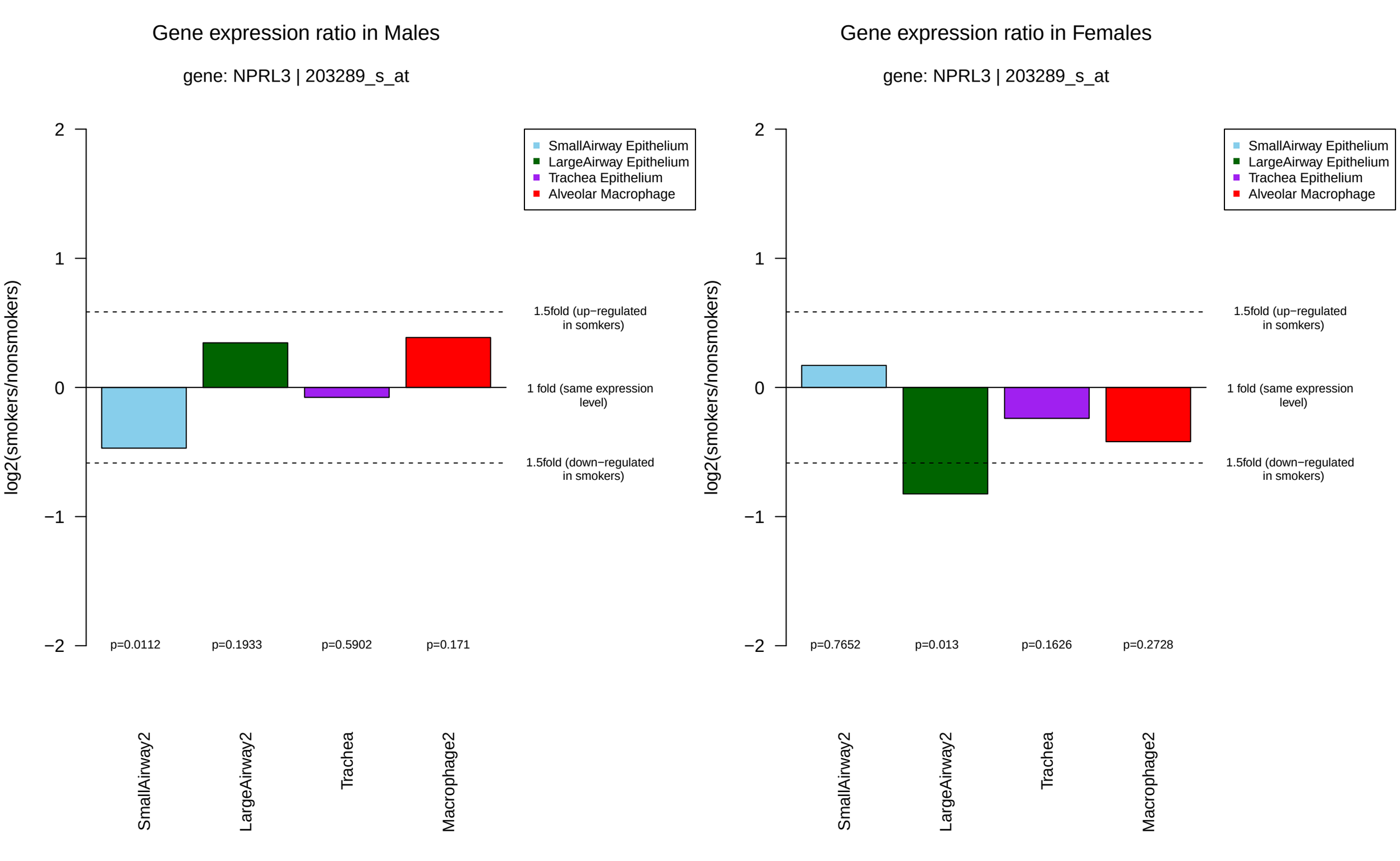
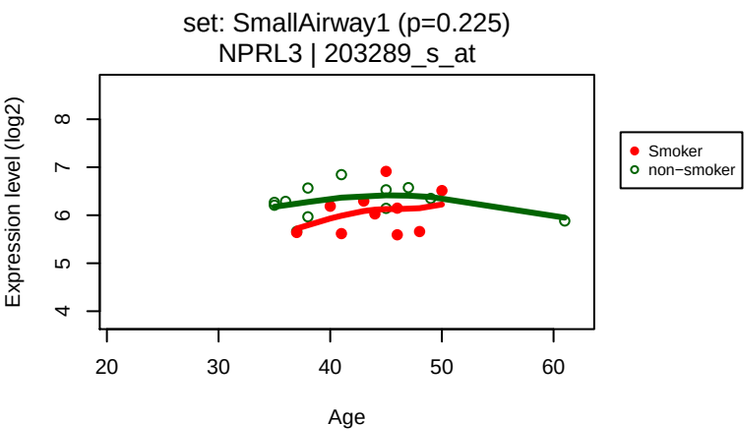
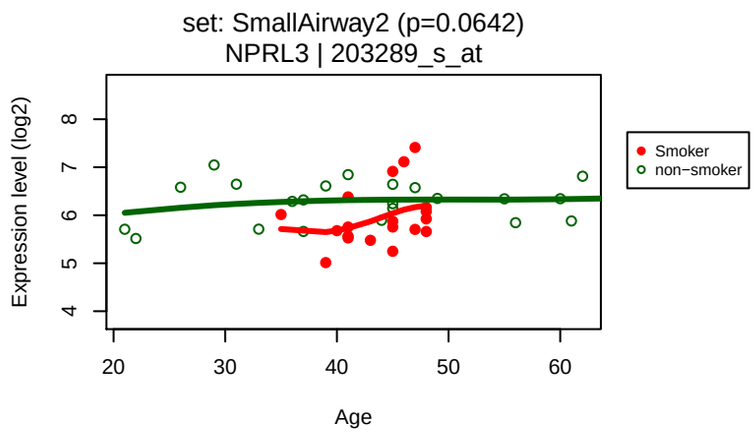
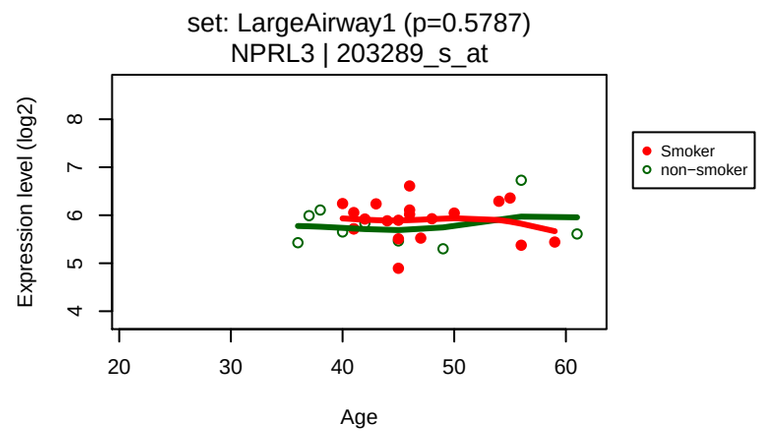

Gene expression of smokers and nonsmokers
gene: NPRL3 | 203289_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.225
p=0.0642
p=0.3827
p=0.5787
p=0.6447
p=0.1448
p=0.7001
p=0.5919
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

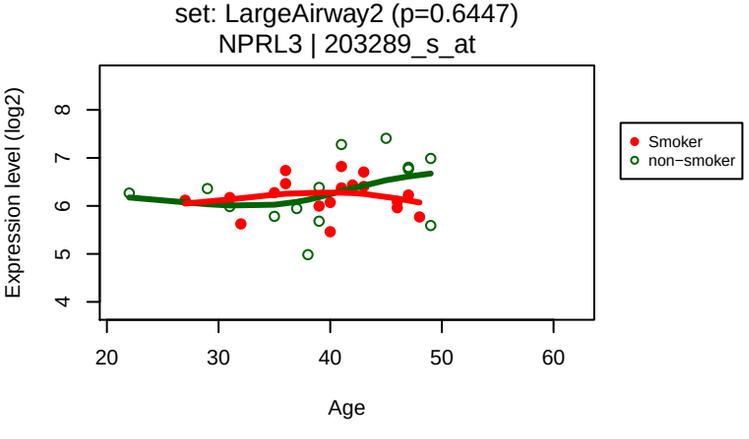

Gene expression ratio (smokers/nonsmokers)
gene: NPRL3 | 203289_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.225
p=0.0642
p=0.3827
p=0.5787
p=0.6447
p=0.1448
p=0.7001
p=0.5919
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

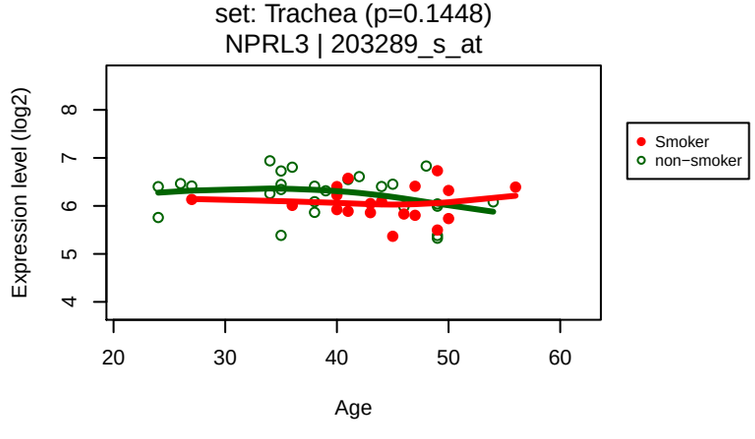

Gene expression ratio in Males
gene: NPRL3 | 203289_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.1933
p=0.5902
p=0.171
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPRL3 | 203289_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7652
p=0.013
p=0.1626
p=0.2728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

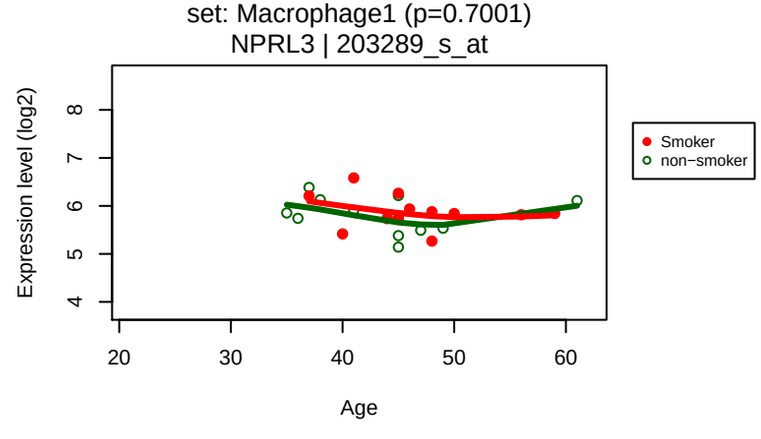

set: SmallAirway1 (p=0.225)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker

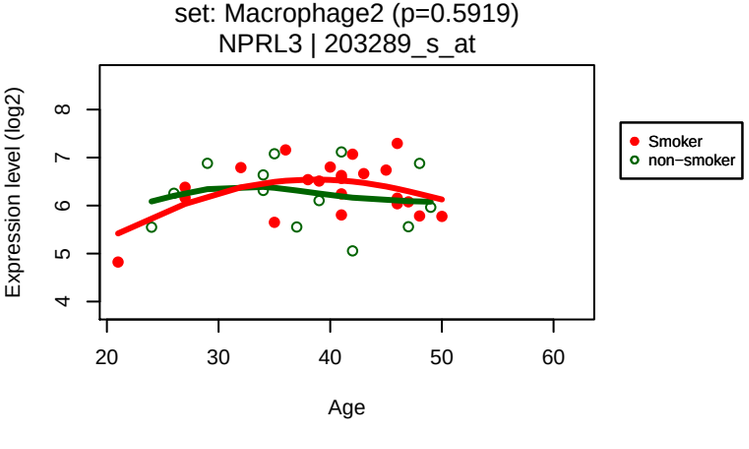

set: SmallAirway2 (p=0.0642)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker

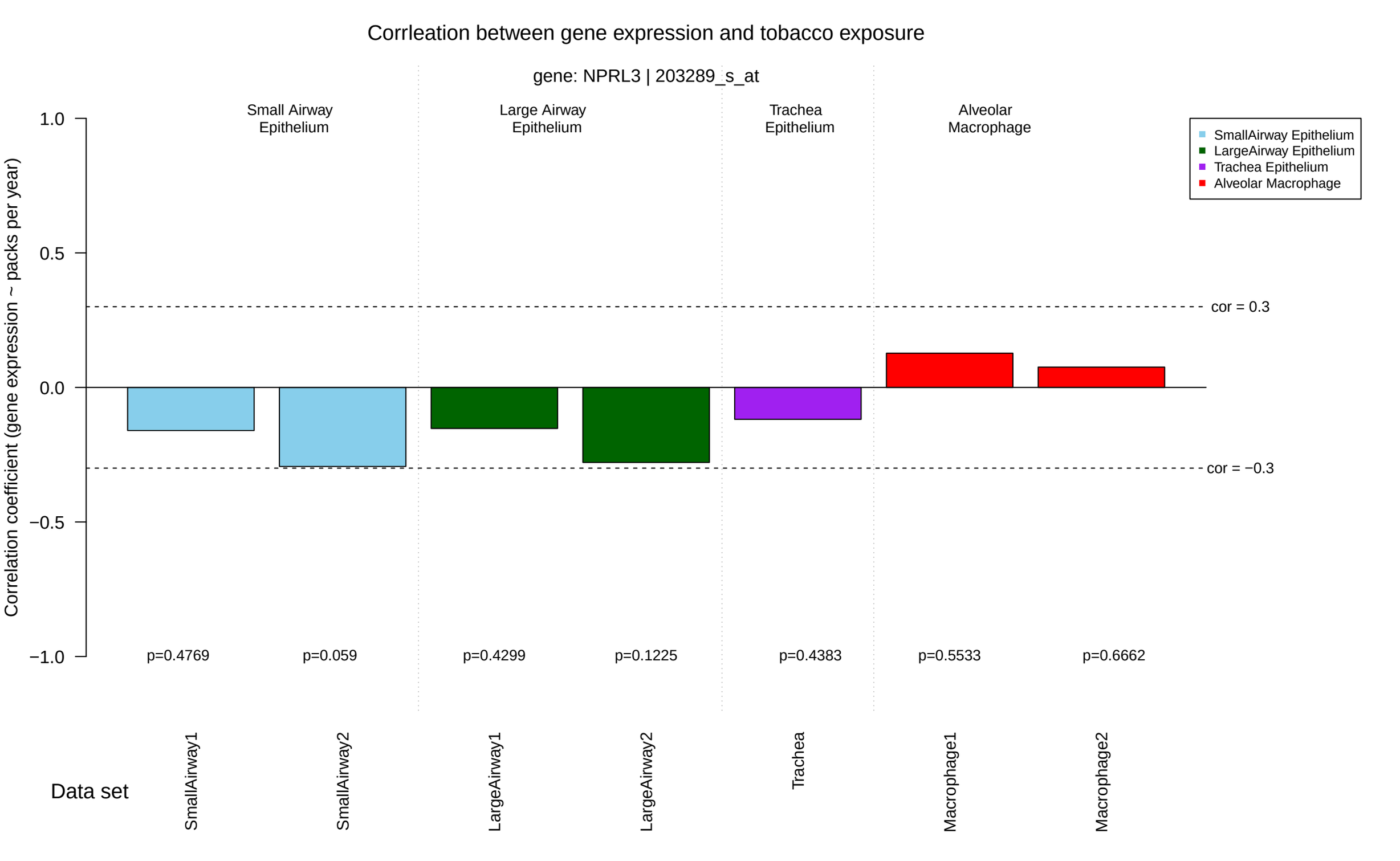

set: LargeAirway1 (p=0.5787)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6447)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1448)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7001)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5919)
NPRL3 | 203289_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPRL3 | 203289_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4769
p=0.059
p=0.4299
p=0.1225
p=0.4383
p=0.5533
p=0.6662
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2